

ERDŐS PROBLEM 928: QUANTITATIVE COMPARISONS AND ORIGINAL FORMULATIONS

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ABSTRACT. We prove explicitly that OpenAI's joint Dickman law implies the density formula $\rho(1/\alpha)\rho(1/\beta)$ in the precise formulation of Erdős Problem 928. Here ρ is the Dickman–de Bruijn function. For fixed $0 < \alpha, \beta < 1$ and real $X \geq 2$, an elementary argument bounds the symmetric difference between the counting sets in the problem and in the cited theorem, restricted to $2 \leq n \leq X$, by $\pi(X^\alpha) + \pi((X+1)^\beta)$, where π counts primes. This bound handles both the strict inequalities and the change in the second threshold from n^β to $(n+1)^\beta$. We also treat the strict formulation with both thresholds based at n , and compare the formulations in Erdős's two 1976 papers and the 1978 Erdős–Pomerance paper. The analytic joint law is due to OpenAI; our contribution consists of the explicit finite comparison estimates and the precise correspondence among these formulations.

1. INTRODUCTION

For an integer $n \geq 2$, let $P(n)$ be the largest prime factor of n , and set $P(1) = 1$. Write ρ for the Dickman–de Bruijn function: the continuous function on $[0, \infty)$ determined by

$$\rho(u) = 1 \quad (0 \leq u \leq 1), \quad u\rho'(u) = -\rho(u-1) \quad (u > 1).$$

The classical smooth-number limit is

$$(1) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{1 \leq n \leq X : P(n) \leq X^c\} = \rho(1/c) \quad (0 < c < 1);$$

see de Bruijn [2, p. 51, (1.1)–(1.2)]. We also use the standard properties $\rho(u) > 0$ for finite $u \geq 0$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$; see [2, p. 52, (1.8)] together with the defining equation. We use $\mathbb{N} = \{1, 2, \dots\}$. All counting variables are integers, and the endpoints X are real. Natural density means the limit of the ordinary, unweighted count up to X , divided by X .

Problem 928 in Bloom's collection [1] asks whether, for fixed $\alpha, \beta \in (0, 1)$, the integers satisfying

$$(2) \quad P(n) < n^\alpha, \quad P(n+1) < (n+1)^\beta$$

have a natural density. The accompanying discussion asks whether this density is $\rho(1/\alpha)\rho(1/\beta)$. The page cites Erdős [3, 4] and Erdős–Pomerance [5]; their formulations will be compared in Section 3.

Our analytic input is OpenAI's joint Dickman law [6, Theorem 1.1, p. 2]: for every fixed $\alpha, \beta \in (0, 1)$,

$$(3) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \leq n^\alpha, P(n+1) \leq n^\beta\} = \rho(1/\alpha)\rho(1/\beta).$$

OpenAI identifies its result as resolving the Erdős–Pomerance joint Dickman conjecture. Section 11.3 of that paper also proves the versions with thresholds X^α, X^β and the corresponding upper-tail conditions. We take these analytic results as given. Proposition 1 supplies explicit elementary estimates for the changes of convention needed in (2) and in Erdős's strict formulation. Section 3 identifies the precise original statements to which the results apply.

2. FINITE COMPARISON ESTIMATES

Fix $\alpha, \beta \in (0, 1)$. For $X \geq 2$, define

$$\mathcal{A}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < (n+1)^\beta\},$$

$$\mathcal{B}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) \leq n^\alpha, P(n+1) \leq n^\beta\},$$

$$\mathcal{C}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < n^\beta\}.$$

The sets $\mathcal{A}$ and $\mathcal{C}$ correspond respectively to Problem 928 and condition (1) in [3, p. 25]; $\mathcal{B}$ is the set in (3). Let $\pi(y) = \#\{p \leq y : p \text{ is prime}\}$, and let $S \triangle T$ denote the symmetric difference of two sets.

Proposition 1 (Comparison of conventions). *For every $\alpha, \beta \in (0, 1)$ and every real $X \geq 2$,*

$$(4) \quad \#(\mathcal{A}_{\alpha, \beta}(X) \triangle \mathcal{B}_{\alpha, \beta}(X)) \leq \pi(X^\alpha) + \pi((X+1)^\beta),$$

$$(5) \quad 0 \leq \#\mathcal{B}_{\alpha, \beta}(X) - \#\mathcal{C}_{\alpha, \beta}(X) \leq \pi(X^\alpha) + \pi(X^\beta).$$

Proof. Suppress the subscripts and the argument X . If $n \in \mathcal{B} \setminus \mathcal{A}$, then $P(n+1) \leq n^\beta < (n+1)^\beta$, so the second condition for $\mathcal{A}$ holds. The first condition must therefore fail, giving $P(n) = n^\alpha$. The prime $p = P(n) \leq X^\alpha$ determines $n = p^{1/\alpha}$ uniquely. Consequently $\#(\mathcal{B} \setminus \mathcal{A}) \leq \pi(X^\alpha)$.

If $n \in \mathcal{A} \setminus \mathcal{B}$, the first condition for $\mathcal{B}$ holds, so

$$n^\beta < P(n+1) < (n+1)^\beta.$$

The intervals $(m^\beta, (m+1)^\beta)$, for integers $m \geq 2$, are pairwise disjoint because $\beta > 0$. Thus $n \mapsto P(n+1)$ is injective on $\mathcal{A} \setminus \mathcal{B}$, and its values are primes less than $(X+1)^\beta$. This proves $\#(\mathcal{A} \setminus \mathcal{B}) \leq \pi((X+1)^\beta)$ and hence (4).

Finally, $\mathcal{C} \subseteq \mathcal{B}$. Any $n \in \mathcal{B} \setminus \mathcal{C}$ satisfies $P(n) = n^\alpha$ or $P(n+1) = n^\beta$. The first equality contributes at most $\pi(X^\alpha)$ integers, by the argument above. For the second, $n = P(n+1)^{1/\beta}$ is determined by a prime at most X^β , giving at most $\pi(X^\beta)$ integers. Summing these two bounds proves (5). $\square$

Corollary 2 (Problem 928 and Erdős's strict formulation). *For every fixed $\alpha, \beta \in (0, 1)$,*

$$(6) \quad \lim_{X \rightarrow \infty} \frac{\#\mathcal{A}_{\alpha, \beta}(X)}{X} = \lim_{X \rightarrow \infty} \frac{\#\mathcal{C}_{\alpha, \beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

In particular, the density in Problem 928 and the density $d(\alpha, \beta)$ asked for in [3, p. 25] both exist and equal the predicted product.

Proof. For finite sets, $|\#S - \#T| \leq \#(S \triangle T)$. Since $\pi(y) \leq [y]$, both upper bounds in Proposition 1 are $O(X^\alpha + X^\beta) = o(X)$. After division by X , equation (3) therefore gives both limits in (6). $\square$

The comparison estimates are independent of the joint Dickman law. They quantify the change of convention for fixed α, β ; the asymptotic evaluation of the count is supplied by (3).

3. COMPARISON WITH THE ORIGINAL SOURCES

We call a threshold *moving* when it depends on the integer being counted, and *fixed* when it is held constant over the counting range. Thus n^α and $(n+1)^\beta$ are moving thresholds, while X^α and X^β are fixed thresholds. Put

$$a(0) = 1, \quad a(t) = 1 - \rho(1/t) \quad (0 < t \leq 1).$$

Table 1 places the reference theorem beside the four source formulations. Its last column records the normalized limits and their justifications. The subsections below explain the changes of convention and counting range.

TABLE 1. Reference theorem and source formulations. In each row, N counts the integers satisfying the listed conditions. Exponents are fixed: $0 < \alpha, \beta < 1$, $\gamma > 1$, and $0 \leq t, s \leq 1$. Here $c_\gamma = \rho(\gamma)$. Unless indicated otherwise, $2 \leq n \leq X$ and $X \rightarrow \infty$.

Formulation and source	Conditions	Limit and justification
Reference theorem OpenAI [6, Thm. 1.1, p. 2]	$P(n) \leq n^\alpha$ $P(n+1) \leq n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ Joint Dickman law (3)
Problem 928 Bloom [1]; Erdős's independence statement [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < (n+1)^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{A}$)
Erdős's strict form Manitoba paper, condition (1) [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{C}$)
Fixed-threshold pair Erdős (Debrecen), $r = 2$ [4, pp. 272–273]	$P(n+1) \leq k$ $P(n+2) \leq k$ $1 \leq n < k^\gamma, k \rightarrow \infty$	$\frac{N}{k^\gamma} \rightarrow \rho(\gamma)^2 = c_\gamma^2$ OpenAI [6, (11.7), p. 80], with the change of variables in Section 3.2
Fixed upper tails Erdős–Pomerance [5, p. 311]	$P(n) \geq X^t$ $P(n+1) \geq X^s$	$\frac{N}{X} \rightarrow a(t)a(s)$ OpenAI [6, (11.8) and following paragraph, p. 80], including equality and endpoints

3.1. Erdős's Manitoba paper. On page 25 of [3], Erdős expects independence for $P(n) < n^\alpha$ and $P(n+1) < (n+1)^\beta$. In the following sentence, he asks whether the integers satisfying his condition (1), namely $P(n) < n^\alpha$ and $P(n+1) < n^\beta$, have a density $d(\alpha, \beta)$. Thus the webpage uses the inequalities in his independence statement, whereas condition (1) uses n^β for the second threshold. Corollary 2 treats both forms explicitly.

Erdős also asks whether these strict inequalities have infinitely many solutions for each fixed pair $\alpha, \beta > 0$. The positive value in (6) implies this, also for parameters outside $(0, 1)$ by choosing smaller positive parameters inside that interval. Infinitude itself was already available: as recorded in [1], it follows from Schinzel [7, Theorem 14, p. 231], applied to $(2n+1)^2 - 1 = 4n(n+1)$. It gives a constant $K > 0$ and infinitely many integers $n \geq 3$ such that

$$P(n(n+1)) \leq n^{K/\log \log n}.$$

Since $K/\log \log n \rightarrow 0$, these integers satisfy Erdős's strict inequalities for every prescribed positive pair α, β , once n is sufficiently large.

3.2. Erdős's Debrecen paper. Write $\gamma > 1$ for the exponent denoted by α in [4, pp. 272–273]. Erdős recalls the one-variable asymptotic $(c_\gamma + o(1))k^\gamma$ for the number of $m \leq k^\gamma$ with $P(m) \leq k$, and conjectures, for each fixed integer $r \geq 1$, that as k tends to infinity through positive integers,

$$(7) \quad \#\{1 \leq n < k^\gamma : P((n+1) \cdots (n+r)) \leq k\} = (c_\gamma^r + o(1))k^\gamma.$$

Equation (1) identifies $c_\gamma = \rho(\gamma)$.

OpenAI [6, (11.7), p. 80] proves, for fixed $c, d \in (0, 1)$,

$$(8) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq m \leq X : P(m) \leq X^c, P(m+1) \leq X^d\} = \rho(1/c)\rho(1/d).$$

For $r = 2$ in (7), put $X = k^\gamma$, $m = n+1$, and $c = d = 1/\gamma$. Since $P(m(m+1)) = \max\{P(m), P(m+1)\}$, the prime-factor conditions coincide with those in (8).

The ranges $1 \leq n < X$ and $2 \leq m \leq X$ differ, after the shift, by at most one endpoint term. Hence (8) gives (7) for $r = 2$.

The webpage [1] states the higher-order conjecture using $P(n+i) \leq n^\alpha$ for $0 \leq i < r$. Its case $r = 2$ is directly (3) with $\beta = \alpha$. The results used here concern two consecutive integers and do not establish the $r \geq 3$ assertion.

3.3. Erdős and Pomerance. On page 311 of [5], the authors use fixed thresholds and upper tails. For $0 \leq t, s \leq 1$, they ask whether

$$(9) \quad b(t, s) = \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \geq X^t, P(n+1) \geq X^s\}$$

exists, is continuous on $[0, 1]^2$, and equals $a(t)a(s)$, where their $a(t)$ is the corresponding one-variable upper-tail limit. Equation (1), together with the treatment of equality and endpoints below, identifies their $a(t)$ with the function defined above.

For $0 < t, s < 1$, inclusion–exclusion applied to (8) and the two marginal laws gives the limit $a(t)a(s)$ with strict upper tails $P(n) > X^t$, $P(n+1) > X^s$. For the shifted marginal, replacing n by $n+1$ changes the counting range by only $O(1)$ terms.

To pass to weak upper tails, observe that for fixed $0 < t \leq 1$ the equality $P(m) = X^t$ is impossible unless X^t is a prime p . In that case $p \mid m$, so there are at most $\lfloor (X+1)/p \rfloor = O(X^{1-t} + 1) = o(X)$ such $m \leq X+1$.

At exponent 0 the upper-tail condition is automatic for $m \geq 2$. At exponent 1, within the range $2 \leq m \leq X+1$, the condition $P(m) \geq X$ forces $m \in [X, X+1]$, so at most two integers occur. This proves (9) at all endpoints as well. Since $\rho(1) = 1$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$, the function a is continuous on $[0, 1]$, and so is $b(t, s) = a(t)a(s)$.

This deduction, including the treatment of equality and endpoints, is already given in [6, (11.8) and following paragraph, p. 80]. The role of the present comparison is to match it to the exact statement in [5]. All four source formulations in Table 1 therefore follow from the cited joint Dickman law and its fixed-threshold version, with the comparisons proved above.

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